Search Strategy and Screening Log

LLM-Based Test Oracles: Source-of-Authority Taxonomy - A Systematic Literature Review

Ali Hassaan Mughal and Muhammad Bilal

Query frozen 2026-05-31; searches executed 31 May 2026

FROZEN SEARCH QUERY v1.0

Frozen: 2026-05-31.

Genre: PRISMA-2020 SLR, Identification stage.

Scope: LLM-based software test oracles (produces an oracle OR empirically studies how the oracle is determined). Window Jan 2021 - May 2026.

Field: Title / Abstract / Keywords.

Concept A - LLM (population)

```
"large language model*" OR LLM OR LLMs OR GPT OR ChatGPT OR Codex OR Copilot
OR Llama OR "Code Llama" OR StarCoder OR Claude OR Gemini OR PaLM
OR "generative AI" OR "foundation model*" OR "code language model"
```

Concept B - Oracle / test (task, oracle-tightened)

```
"test oracle*" OR "oracle generation" OR "oracle inference" OR "oracle problem"
OR "assertion generation" OR "test assertion*" OR "exception oracle"
OR "metamorphic testing" OR "metamorphic relation*" OR "property-based testing"
OR "differential testing" OR postcondition* OR precondition*
OR "user acceptance test*" OR "acceptance test*" OR Gherkin
OR "behavior-driven" OR "behaviour-driven" OR "behavior-driven development"
```

Search = A AND B

Per-database renderings (syntax-adapted only; logic identical)

Scopus (author runs; CS subject restriction handles cross-domain noise)

```
TITLE-ABS-KEY("large language model*" OR LLM OR LLMs OR GPT OR ChatGPT OR Codex OR Copilot OR Llama
     OR "Code Llama" OR StarCoder OR Claude OR Gemini OR PaLM OR "generative AI" OR "foundation
    model*" OR "code language model")
AND TITLE-ABS-KEY("test oracle*" OR "oracle generation" OR "oracle inference" OR "oracle problem"
    OR "assertion generation" OR "test assertion*" OR "exception oracle" OR "metamorphic testing"
    OR "metamorphic relation*" OR "property-based testing" OR "differential testing" OR
    postcondition* OR precondition* OR "user acceptance test*" OR "acceptance test*" OR Gherkin OR
    "behavior-driven" OR "behaviour-driven" OR "behavior-driven development")
AND PUBYEAR > 2020 AND PUBYEAR < 2027
AND SUBJAREA(COMP)
```

IEEE Xplore (CS-native; All-Metadata field; year 2021-2026 in UI)

Concept A AND Concept B over "All Metadata", year range 2021-2026.

ACM Digital Library (CS-native; Abstract field; year 2021-2026)

Concept A AND Concept B over Abstract, E-Pub 2021-2026.

Refinement rationale vs prior string

- Added Scopus SUBJAREA(COMP): removes palm-oil / Codex-Alimentarius / gherkin-pickle / food-sensory false positives WITHOUT deleting terms (preserves recall).
- Wildcards/plurals + added oracle-relevant concepts (oracle problem, oracle inference, precondition*) for recall.
- "metamorphic" kept only as 2-word phrases (drops geology noise).
- Dropped standalone "BDD" (collides with Binary Decision Diagram); kept spelled-out behavior-driven forms.
- Oracle-tight: no generic "unit test" / "test generation" (those are screened, not searched).

Run ledger (REAL engine-reported counts)

Database	Date	Field	Reported count	Notes
Scopus	2026-05-31	TITLE-ABS-KEY + SUBJAREA(COMP)	**391**	Documents tab; CS subject facet = 391 (all hits CS). Prior string gave 301; +90 from added recall terms. Preprints tab separate (not counted).
IEEE Xplore	2026-05-31	All Metadata	**234**	2021-2026 filter applied (262 before year filter). Conf 199 / Jrnl 27 / EarlyAccess 6 / Mag 2. All-Metadata field = high recall (prior Abstract-only run = 71). matchBoolean=true.
ACM DL	2026-05-31	Abstract (both concepts)	**1,811**	ACM Full-Text Collection (847,884 records); E-Pub 01/01/2021-12/31/2026. Two Abstract rows AND-joined. HIGH vs Scopus/IEEE: generic recall terms (precondition/postcondition/acceptance test/differential/property-based/metamorphic testing) are common in ACM CS abstracts -> high recall, low precision; screening will cut heavily. Prior tighter protocol run = 38.

Identification subtotal (pre-dedup): 391 + 234 + 1,811 = 2,436.

Default sort order at search time (display order only; does NOT affect set membership or counts, since every export is the full result set, not a top-N)

Database	Default sort used	Notes
Scopus	**Date (newest first)**	Scopus advanced-search results default.
IEEE Xplore	**Relevance**	IEEE results default ("Sort By: Relevance").
ACM DL	**Recency**	ACM results default ("Recency").

Recorded per reproducibility (PRISMA). All three exports = "all results", so sort does not change which records are captured.

Record capture status (full SLR: capture ALL records, no caps)

Database	Records	File	Status
IEEE Xplore	234	the `Search-IEEE` sheet	DONE (single CSV export)
ACM DL	1,811	the `Search-ACM` sheet	DONE via 3 non-overlapping date-split BibTeX exports (ACM export caps at 1,000): 2021-2024=652, 2025=720, 2026=439 -> merged, 1,811 unique keys.
Scopus	391	the `Search-Scopus` sheet	DONE (single CSV export, 391 records, full fields incl. DOI; abstracts are removed from the released exports).

STAGE 1 (IDENTIFICATION) COMPLETE: 2,436 records captured across 3 per-platform files (Scopus 391 + IEEE 234 + ACM 1,811). Next PRISMA stage = deduplication, then title/abstract screening against pre-registered inclusion/exclusion criteria.

Note on ACM date-split per-year counts (these are real per-year identification numbers): 2021-2024=652, 2025=720, 2026=439.

Search log

Window: January 2021 to May 2026. **Field:** Title / Abstract / Keywords. **Query:** Concept A (LLM terms) AND Concept B (oracle/test terms), frozen v1.0 on 2026-05-31. Per-database result exports are the `Search-Scopus`, `Search-IEEE`, and `Search-ACM` sheets of `Review-Data.xlsx` (metadata only; abstracts removed for copyright).

Run ledger (engine-reported counts, 2026-05-31)

Database	Field / filter	Reported count	Released export
Scopus	TITLE-ABS-KEY + SUBJAREA(COMP), PUBYEAR 2021-2026	391	Search-Scopus
IEEE Xplore	All Metadata, 2021-2026	234	Search-IEEE
ACM DL	Abstract (A AND B), E-Pub 2021-2026	1,811	Search-ACM
Total identified (pre-dedup)		2,436	

ACM export hard-caps at 1,000, so ACM was captured in three non-overlapping date-split batches and merged (1,811 unique keys): 2021-2024 = 652, 2025 = 720, 2026 = 439.

PRISMA stages

- **Identification.** 2,436 records captured across the three databases (complete result sets, no caps).
- **De-duplication.** 2,436 to 2,245 unique records (191 duplicates removed; match key normalized DOI, else normalized title).
- **Screening.** An LLM recall-oriented pre-filter reduced the 2,245 records to 178 retained (79 include, 99 maybe). Two authors then independently screened a 385-record verification set (all 178 retained plus a fixed-seed, reason-stratified 10% sample, 207 records, of the excluded set), blind to the model labels and to each other, recording an Include, Maybe, or Exclude decision with a rationale for every record. Independent dual-screen agreement was Cohen's kappa = 0.79. The adjudicated set taken to full text was 115 records. Reviewer decisions are in the `Decisions-Ali`, `Decisions-Bilal`, and `Decisions-Consensus` sheets; every record and its decision are in `Screening-Funnel`.
- **Eligibility.** Full-text assessment of the 115 records against the locked criteria yielded 54 included studies.

Reproducibility notes

- All three exports were complete result sets (no top-N truncation), so the display sort order affects display only, not set membership.
- Per-database syntax differs (adapted per platform); the underlying Boolean logic is identical (see the query above).